Exp9

>> git clone https://github.com/HumayunK01/machine-observing

>> cd machine-observing

Run npm install prom-client express

Step 3: Run the Stack via Docker Compose (Step f in manual)

> [!IMPORTANT]

> Make sure **Docker Desktop** is running in Windows before this step. (If you don't have Docker installed/running, see the **No-Docker Backup Method** below).

In the terminal inside `machine-observing`, run:

```powershell
docker-compose up -d
```

You will see Docker pull the images and show:

```text
[+] Running 5/5
✓ node-app Built
✓ Network machine-observing_default Created
✓ Container machine-observing-node-app-1 Started
✓ Container machine-observing-prometheus-1 Started
✓ Container machine-observing-grafana-1 Started
```

Step 4: Verify Prometheus (Page 5 in manual)

1. Open your browser and go to:

   ```
   http://localhost:9090
   ```

2. In the query bar, enter:

   ```promql
   process_cpu_seconds_total{job="node-app"}
   ```

   or:

   ```promql
   http_requests_total
   ```

3. Click **Execute** and switch to the **Graph** tab. Take a screenshot here (matches Page 5).

Step 5: Setup Grafana Dashboard (Page 5 & 6 in manual)

1. Open your browser and go to:

   ```
   http://localhost:3001
   ```

2. Login with credentials:

   - **Username:** `admin`
   - **Password:** `admin`

3. Add Prometheus as a Data Source:

   - Go to **Connections** > **Data Sources** > **Add data source**.
   - Select **Prometheus**.
   - In Prometheus server URL, set:

     ```
     http://prometheus:9090
     ```

(If running outside Docker on localhost, use `http://localhost:9090`).

- Click **Save & Test**.

4. Import / Create Dashboard:

- Create a dashboard or import Dashboard ID `11159` (Node.js Application Dashboard).

- This produces the exact graphs shown on Page 6 of your manual.

What if Docker is NOT working or not installed? (Quick Standalone Method)

If Docker Desktop is not running on your PC during the exam, you can run Prometheus directly on Windows:

1. **Download Prometheus for Windows:**

- Go to prometheus.io/download

- Under Prometheus, download the **Windows amd64** `.zip` file (e.g. `prometheus-*.windows-amd64.zip`).

- Extract the zip folder. Inside you will find `prometheus.exe`.

2. **Start your Node.js application:**

In terminal:

```powershell
node server.js
```

(Running at `http://localhost:3000` with metrics at `http://localhost:3000/metrics`)

3. **In `prometheus.yml`, change target to localhost:**

```yaml
targets: ["localhost:3000"]
```

4. **Run Prometheus:**

```powershell
.\prometheus.exe --config.file=prometheus.yml
```

Prometheus is now live at `http://localhost:9090`.

Step 3: Open Prometheus & Take Lab Screenshots

1. **Check targets are UP:**
    - Open your browser and visit: **http://localhost:9090/targets**
    - You will see node-app (1/1 up) with a green **UP** status indicator.
2. **Query metrics (matches Page 5 of your manual):**
    - Go to: **http://localhost:9090**
    - Click the **Graph** tab.
    - In the search bar, type:

promql

process_cpu_seconds_total{job="node-app"}

    - Click **Execute**. You will see the time-series graph updating every 5 seconds

3. Generate some Traffic (To see a rising line)

To make the graph move up like in the manual:

1. Open a new tab and go to **http://localhost:3000/**
2. Refresh that page 10–15 times (F5).
3. Now back in Prometheus, you can also query:

promql

http_requests_total

Click **Execute** and look at the **Graph** tab — you will see the line climb with each request you made!